

Descend, and plot the path

You did five steps on paper. Now turn them into a function, draw the walk down the curve, and find the cliff edge of the learning rate yourself instead of taking my word for where it is.

Scaffolded lab

16 minutes

Keep your numgrad

Last lab of unit 1

1 · descend 2 · the path 3 · the loss curve 4 · six trajectories 5 · find the cliff
6 · a different function Keep numgrad If you finish early Reflection

TASK 1 · 3 MINUTES

Write descend(x0, eta, steps)

1

Open the provided `lab06.ipynb`. The skeleton below is already in it. `descend` returns the full list of x values visited, **starting with x_0** , so five steps gives six numbers. The update is the one line from part 2 of the lecture.

the skeleton – three lines are yours

PYTHON

```
import numpy as np
import matplotlib.pyplot as plt

def f(x):      return x ** 2
def fprime(x): return 2 * x

def descend(x0, eta, steps):
    xs = [x0]
    x = x0
    # your loop here: update x, then append it
    return xs
```

check it against your paper

PYTHON

```
print([round(v, 4) for v in descend(2.0, 0.1, 5)])  
# [2.0, 1.6, 1.28, 1.024, 0.8192, 0.6554]
```



Digit for digit against your paper table

If your code and your paper disagree, do not assume the code is right. One of the two has an error and finding out which is the point of the task. The most common cause is appending before updating, which gives you the starting value twice.

TASK 2 · 3 MINUTES

Plot the path on the curve

2

- a Draw $f(x) = x^2$ as a line, using `np.linspace(-2.5, 2.5, 200)`.
- b Scatter the visited points on top of it — you need both `xs` and `f(xs)`.
- c Label both axes, add a legend naming the curve and the path.

You should see the steps bunching up as they approach zero. Write one sentence saying why they do — the answer is in the update rule, not in the picture.

TASK 3 · 2 MINUTES

The loss curve, and the log axis

3

A second figure: $f(x)$ against step number, for 30 steps at $\eta = 0.1$. Then add one line:

PYTHON

```
plt.yscale("log")
```

Look again, and write one sentence on what the log axis shows you that the linear one hid. This is the most useful plot in the course and you will draw it in almost every session from 10 to 26.

TASK 4 · 4 MINUTES · THE MAIN TASK

Six trajectories, and a formula

4

Run $\eta = 0.01, 0.1, 0.5, 0.9, 1.0, 1.1$ for 8 steps each from $x_0 = 2$, and print the table from part 5 of the lecture.

Then verify numerically that every trajectory is exactly `$x_0 * (1 - 2*\eta)^n$` :

PYTHON

```
for eta in [0.01, 0.1, 0.5, 0.9, 1.0, 1.1]:
    xs = descend(2.0, eta, 8)
    closed = [2.0 * (1 - 2*eta) ** n for n in range(9)]
    print(eta, np.allclose(xs, closed), round(abs(1 - 2*eta), 3))
```

**The third column is the whole explanation**

Each step multiplies x by `$(1 - 2\eta)$` . Everything you observe follows from whether the size of that number is below one, equal to one, or above it. At $\eta = 0.5$ it is exactly zero, which is why that run reaches the answer in a single step — and why "bigger learning rate is faster" stops being true at exactly the halfway point.

TASK 5 · 2 MINUTES

Find the cliff yourself

5

Search for the largest η that still converges, to two decimal places. State it, then explain it in one sentence using `$|1 - 2\eta|$` .

**Give it enough steps, or you will find the wrong cliff**

At $\eta = 0.99$ the factor is 0.98, and 0.98 to the power of 200 is still 0.018 — so a 200-step test reports that 0.99 diverges, which is false. Run at least 2000 steps before you decide, and treat `inf` or `nan` as "did not converge" rather than as an error. That slowness is part of the answer to this task, not an obstacle to it.

$\eta = 0.99$. Convergence needs $|1 - 2\eta| < 1$, which holds for every η strictly between 0 and 1. At $\eta = 1.0$ the factor is exactly -1 , so x flips sign and never shrinks — it oscillates between $+2$ and -2 forever. Above 1.0 it grows.

Worth noticing: 0.99 converges, but appallingly slowly, because 0.98 to the power of eight is still 0.85. The largest learning rate that *works* and the best one to *use* are nowhere near each other, which is the practical lesson of the whole session.

TASK 6 · 2 MINUTES

A different function

6

Now minimise $g(x) = (x - 3)^2 + 1$. Write `gprime`, run `descend` from $x_0 = 0$ with $\eta = 0.1$, and confirm it heads for 3 rather than 0.

- a Where does it converge to, and what is the loss there? Neither answer is zero.
- b Does the same cliff at $\eta = 1.0$ apply? Say why in one sentence — the shape of the curve has not changed, only where it sits.



The point of task 6

The minimum is not at zero and the loss at the minimum is not zero. Both of those are true of every real model you will train — a trained network has residual error, and the parameters that produce it are not special round numbers. Session 10 fits a line to six noisy points and lands in exactly this situation.

Keep numgrad

If you wrote a numerical-gradient checker in part 6 of the lecture, save it in a cell of its own and label it clearly.

the one piece of unit 1 code that comes with you

PYTHON

```
def numgrad(f, x, h=1e-5):  
    return (f(x + h) - f(x - h)) / (2 * h)  
  
print(numgrad(f, 2.0), fprime(2.0))    # 4.0 and 4.0
```

In session 14 you will derive gradients for a whole network by hand and then need to know whether your derivation is right. This four-line function is how you will find out, and it is the reason that session is

survivable.

If you finish early

Unmarked. One extension, and it is worth the time.

Descend in two dimensions. Minimise $f(x, y) = x^2 + 2y^2$. Its gradient is $(2x, 4y)$, so one step updates both coordinates at once: $x \leftarrow x - \eta \cdot 2x$ and $y \leftarrow y - \eta \cdot 4y$.

- a Take 20 steps from $(4, 4)$ with $\eta = 0.1$ and plot the path in the x - y plane. A contour plot of f behind it (`plt.contour`) is a bonus.
- b Which coordinate reaches zero first? Explain it with the two factors $|1 - 2\eta|$ and $|1 - 4\eta|$ — the idea from task 4, applied once per direction.
- c Find the new cliff. It is *not* at $\eta = 1.0$ any more. Say in one sentence why the steepest direction sets the limit for the whole walk.

(b) At $\eta = 0.1$ the x factor is 0.8 and the y factor is 0.6, so y shrinks faster: after 20 steps y is about 0.0001 while x is still about 0.046. The path bends towards the x -axis and then crawls along it.

(c) $\eta = 0.5$. Both factors must be below one in size, and $|1 - 4\eta| < 1$ is the stricter condition. At $\eta = 0.5$ the y factor is exactly -1 , so y bounces between $+4$ and -4 forever while x has already arrived. One learning rate has to suit every direction at once — the first glimpse of a problem you will meet again in session 19.

Reflection · end of unit 1



Write it in the notebook, in your own words

Connect the two big ideas of this unit in three or four sentences: *a matrix multiply computes a model's prediction* (Lab 5) and *gradient descent changes the numbers inside that model* (this lab). What, exactly, would gradient descent be changing in the 2–3–1 network you built on Tuesday?